

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

Branch/ Semester	BCA SEM - 1 (AY2026-27)
Class	B
Subject Code	05BC03103
Subject Name	Programming Practices - 1 (PP1)
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Sr. No.	Code Efficiency Review [10] Find out the output of below C program code	Answer Output
1	<pre>int main() { int a = 5; int b = 10; printf("%d", a + b * b); return 0; }</pre>	105
2	<pre>int main() { int x = 20; int y = 5; printf("%d", x / y + x * y); return 0; }</pre>	104
3	<pre>int main() { printf("Programming\n"); printf("Practice\n"); printf("1"); return 0; }</pre>	Programming Practice 1
4	<pre>int main() { int a = 7; printf("%d", a * a * a); return 0; }</pre>	343

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

	}	
5	<pre>#define PI 3.14 int main() { float radius = 2; printf("%.2f", PI * radius * radius); return 0; }</pre>	12.56
6	<pre>int main() { char ch = 'A'; printf("%d", ch + 5); return 0; }</pre>	70
7	<pre>int main() { int a = 10; float b = 3; printf("%.2f", a / b); return 0; }</pre>	3.33
8	<pre>int main() { int x = 5; const int y = 10; printf("%d", x + y); return 0; }</pre>	15
9	<pre>int main() { int a = 5; int b = 2; printf("%d", a + b * a - b); }</pre>	13

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	<pre> return 0; } </pre>	
10	<pre> int main() { int x = 10; printf("%d %d", ++x, x++); return 0; } </pre>	12 10
11	<pre> int main() { int a = 12; int b = 5; printf("%d", a % b + a / b); return 0; } </pre>	4
12	<pre> int main() { int a = 5; printf("%d", a > 3 && a < 10); return 0; } </pre>	1
13	<pre> int main() { int x = 15; if (x > 10) { printf("A"); } else { printf("B"); } return 0; } </pre>	A

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14	<pre> int main() { int i; for (i = 1; i <= 5; i++) { printf("%d ", i * i); } return 0; } </pre>	1 4 9 16 25
15	<pre> int main() { int i = 1; while (i <= 5) { if (i == 3) { break; } printf("%d ", i); i++; } } </pre>	1 2
16	<pre> int main() { int n = 2; switch (n) { case 1: printf("One"); break; case 2: printf("Two"); case 3: printf("Three"); break; default: printf("Default"); } } </pre>	TwoThree

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Bachelor of Computer Applications (BCA)

	<pre> return 0; } </pre>	
17	<pre> int square(int x) { return x * x; } int main() { printf("%d", square(4)); return 0; } </pre>	16
18	<pre> int fun(int n) { if (n == 0) { return 0; } return n + fun(n - 1); } int main() { printf("%d", fun(4)); return 0; } </pre>	10
19	<pre> #define SQUARE(x) x * x int main() { printf("%d", SQUARE(3 + 1)); return 0; } </pre>	7
20	<pre> int factorial(int n) { if (n == 1) { return 1; } } </pre>	120

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	<pre> return n * factorial(n - 1); } int main() { printf("%d", factorial(5)); return 0; } </pre>	
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Sr. No.	Debug Hunt: [10] Find out the Errors from below C program code and write the correct code.	CO	Bloom's Level
1	<pre> int Main() { int num = 10 Printf("Number = %d\n", num); return; } </pre>	CO1	Analyze (C4)
Ans:1	Error Description : main.c: In function 'Main': error: expected ',' or ';' before 'Printf' error: 'return' with no value, in function returning non-void Correct Code : <pre> Int main() Int num=10; Printf("Number=%d\n",num); return 0; </pre>		
2	<pre> int main() { int a = 20; int b = 30; printf("Sum = %d\n", a + c); } </pre>	CO1	Analyze (C4)

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	<pre>return 0; }</pre>		
Ans:2	Error Description : error: 'c' undeclared (first use in this function) Correct Code : <pre>Printf("Sum=%d\n",a+b);</pre>		
3	<pre>int main({ printf("Programming in C); return 0; }</pre>	CO1	Analyze (C4)
Ans:3	Error Description : error: expected declaration specifiers or '...' before '{' token warning: missing terminating " character Correct Code : <pre>Int main() Printf("Programming in C");</pre>		
4	<pre>int main() { integer marks = 80; printf("%d", marks) return 0; }</pre>	CO1	Evaluate (C5)
Ans:4	Error Description : error: unknown type name 'integer' error: expected ';' before 'return' Correct Code : <pre>int marks=80; printf("%d",marks);</pre>		
5	<pre>int main() { float total marks = 75.5; printf("%f", total marks); return 0; }</pre>	CO2	Analyze (C4)
Ans:5	Error Description : <pre>main.c:12:17: error: expected '=', ',', ';', 'asm' or '__attribute__' before 'marks'</pre>		

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

	error: 'marks' undeclared (first use in this function) error: 'total' undeclared (first use in this function) Correct Code : float total_marks=75.5; printf("%f",total_marks);		
6	<pre>#define PI = 3.14 int main() { float area; int r = 5; area = PI * r * r; printf("%.2f", area); return 0; }</pre>	CO2	Evaluate (C5)
Ans:6	Error Description : error: expected expression before '=' token #define PI = 3.14 note: in expansion of macro 'PI' Correct Code : #define PI 3.14		
7	<pre>int main() { const int value; value = 50; printf("%d", value); return 0; }</pre>	CO2	Evaluate (C5)
Ans:7	Error Description : error: assignment of read-only variable 'value' Correct Code : const int value=50;		
8	<pre>int main() { char grade = "A"; float percentage = 85;</pre>	CO2	Analyze (C4)

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	<pre>printf("%c %.2f", grade, percentage); return 0; }</pre>		
Ans:8	Error Description : error: initialization of 'char' from 'char *' makes integer from pointer without a cast Correct Code : Char grade='A';		
9	<pre>int main() { int a = 10; if (a = 20) printf("Equal"); return 0; }</pre>	CO3	Evaluate (C5)
Ans:9	Error Description : warning: suggest parentheses around assignment used as truth value [-Wparentheses] if (a = 20) ^ Correct Code : If(a==20)		
10	<pre>int main() { int age; scanf("%d", age); printf("%f", age); return 0; }</pre>	CO3	Analyze (C4)
Ans:10	Error Description : error: implicit declaration of function 'scanf' [-Wimplicit-function-declaration] scanf("%d", age); warning: format '%d' expects argument of type 'int *', but argument 2 has type 'int' [-Wformat=] scanf("%d", age); error: implicit declaration of function 'printf' [-Wimplicit-function-declaration] printf("%f", age); warning: format '%f' expects argument of type 'double', but argument 2 has type 'int' [-Wformat=] printf("%f", age); Correct Code : Scanf("%d",&age);		

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

	<code>printf("%f",age);</code>		
11	<pre>int main() { int a = 5; int b = 10; printf("%d", a++ + ++a + b--); return 0; }</pre>	CO3	Evaluate (C5)
Ans:11	Error Description : error: implicit declaration of function 'printf' warning: operation on 'a' may be undefined Correct Code : <code>int x = a++; int y = ++a;</code> <code>printf("%d", x + y + b--);</code>		
12	<pre>int main() { int x = 10; int y = 3; printf("%d", x / y * y == x); return 0; }</pre>	CO3	Analyze (C4)
Ans:12	Error Description : main.c: In function 'main': main.c:1:34: warning: implicit declaration of function 'printf' [-Wimplicit-function-declaration] <code>int main() { int x = 10; int y = 3; printf("%d", x / y * y == x); return 0; }</code> Correct Code : <code>double x = 10.0;</code> <code>double y = 3.0;</code>		
13	<pre>int main() { int i; for(i = 1; i <= 5; i--) printf("%d ", i); return 0; }</pre>	CO4	Analyze (C4)

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

Ans:13	Error Description : It will turn into infinite loop Correct Code : <pre>For(i=1;i<=5,i++)</pre>		
14	<pre>int main() { int i = 1; while(i <= 10) { printf("%d ", i); } return 0; }</pre>	CO4	Evaluate (C5)
Ans:14	Error Description : Its infinite loop prints 1s Correct Code : <pre>printf("%d ", i); i++;</pre>		
15	<pre>int main() { int choice = 2; switch(choice) { case 1: printf("One"); case 2 printf("Two"); default: printf("Invalid"); } return 0; }</pre>	CO4	Evaluate (C5)
Ans:15	Error Description : error: expected ':' before 'printf' Correct Code : <pre>Case 2:</pre>		

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

16	<pre>int main() { int marks = 85; if(marks >= 90) printf("A"); else if(marks >= 80); printf("B"); else printf("C"); return 0; }</pre>	CO4	Evaluate (C5)
Ans:16	Error Description : error: 'else' without a previous 'if' Correct Code : <pre>if(marks >= 90) { printf("A"); } else if(marks >= 80) { printf("B"); } else { printf("C"); }</pre>		
17	<pre>factorial(int n) { if(n == 1) return 1; return n * factorial(n-1); } int main() { printf("%d", factorial()); return 0; }</pre> ``` CO-5 Evaluate (C5)	CO5	Evaluate (C5)
Ans:17	Error Description : warning: return type defaults to 'int' error: too few arguments to function 'factorial' Correct Code : <pre>int factorial(int n)</pre>		

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

	<pre>if(n <= 1) printf("%d\n", factorial(5));</pre>		
18	<pre>int sum(int n) { return n + sum(n-1); } int main() { printf("%d", sum(5)); return 0; }</pre>	CO5	Analyze (C4)
Ans:18	Error Description : Process returned -1073741571 (0xC00000FD) [This is the Windows hex code for Stack Overflow] Correct Code : <pre>if (n <= 0) { return 0; } return n + sum(n - 1);</pre>		
19	<pre>int main() { printf("%d", SQUARE(2+3)); return 0; }</pre>	CO5	Evaluate (C5)
Ans:19	Error Description : error: 'SQUARE' undeclared (first use in this function) Correct Code : <pre>#define SQUARE(x) ((x) * (x))</pre>		
20	<pre>int add(int, int); void main() { printf("%d", add(10)); return 0; } int add(int a, int b) { return a + b; }</pre>	CO5	Evaluate (C5)

Faculty of Computer Applications (FCA)

Bachelor of Computer Applications (BCA)

Ans:20	Error Description : warning: implicit declaration of function 'printf' error: too few arguments to function 'add' warning: 'return' with a value Correct Code : <code>printf("%d\n", add(10, 5));</code>
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Submission Guidelines:

- Write the answers of above questions at appropriate place **without using AI Tool**.
- In Error finding questions, **highlight the error statement with RED** colour in question and **correct statement with BLUE** Colour in answer.
- Create a pdf and give the name your enrolment no – Name
- **Upload this pdf in Google Class room on or before 24/10/2026.**
- Marks will be given based on originality of the content and submission on time.